

Backward Reasoning in Isabelle/HOL

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theory BW
imports Main
begin

lemma implication_introduction:
  " $X \implies (Y \longrightarrow X)$ "
  sorry

lemma neg_intro:
  " $(X \longrightarrow Y) \implies (X \longrightarrow \neg Y) \implies \neg X$ "
  sorry

lemma assumes
  assum1: " $P \longrightarrow Q$ " and assum2: " $Q \longrightarrow R$ " and assum3: " $\neg R$ "
  shows " $\neg P$ "
proof-

  have 1: " $\neg R$ "
    using assum3
    .
  have 2: " $Q \longrightarrow R$ "
    using assum2
    .
  have 3: " $Q \longrightarrow \neg R$ "
    apply(rule implication_introduction)
    using 1
    .
  have 4: " $\neg Q$ "
    apply(rule neg_intro)
    using 2 3
    .

  have 5: " $P \longrightarrow \neg Q$ "
    apply(rule implication_introduction)
    using 4
    .

  show " $\neg P$ "
    apply(rule neg_intro)
    using assum1 5
    .

qed

end
```